

VEDANT PURI

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OBJECTIVE

PhD candidate graduating August 2026 seeking a technical role in machine learning research. Experience developing scalable-transformers and custom GPU kernels for large-scale training. Background in computational fluid dynamics.

EDUCATION

Carnegie Mellon University Jan 2022–Present

Ph.D Mechanical Engineering. Advisors: Levent Burak Kara, Yongjie Jessica Zhang

Proposed thesis: *Neural representations for computational physics: from reduced order modeling to transformers*

University of Illinois Urbana-Champaign 2015–2019

B.S. Engineering Mechanics, B.S. Mathematics.

SELECTED RESEARCH CONTRIBUTIONS

FLARE: Fast Low-Rank Attention Routing Engine | *Efficient attention architectures* 2025

- Derived a flexible low-rank reformulation of self-attention via latent routing.
- Demonstrated scaling to 1M tokens on a single H100 GPU, attaining over 200× speedup over vanilla self-attention.
- Integrated into NVIDIA PhysicsNeMo as an experimental model for scientific machine learning workloads.

FLARE for Language Modeling (Ongoing dissertation work) | *Decoder-only architectures* 2025–Present

- Extending FLARE to decoder-only next-token prediction with causal attention.
- Enabling adaptive attention state size to control memory and compute during training and inference.
- Implementing fused Triton kernels for causal training, prefill, and decode.

Equation-based PDE modeling with neural manifolds | *Hybrid data + physics methods* 2024

- Introduced smooth neural fields as nonlinear spatial ansatz functions in equation-based reduced-order modeling.
- Retained physics-based Galerkin time evolution while learning expressive low-dimensional representations.
- Attained 199× speedup over full-order simulations in transport-dominated regimes.

EXPERIENCE

Carnegie Mellon University | *Research Assistant* Jan 2022–Present

- Wrote solvers for phase field simulations of lithium dendritic growth in solid-state batteries.
- Wrote fluids dynamics solvers for turbulence closure modeling with differentiable physics.

Julia Computing | *Intern Engineer* Apr 2021–Nov 2021

- Built numerical solvers for scientific machine learning ecosystem in Julia.

Carnegie Mellon University | *Research Assistant* Sep 2020–Jan 2021

- Developed differentiable geometry representations and meshing algorithms.

Argonne National Laboratory | *Research Assistant* Mar 2020–Sep 2020

- Executed large-scale simulations of turbulent airflow over urban landscapes on supercomputers.

Argonne National Laboratory | *Research Assistant* May 2018–Jul 2018

- Executed high-fidelity fluid dynamics simulations and analyzed turbulent statistics for closure modeling.

National Center for Supercomputing Applications | *Intern* Sep 2017–May 2018

- Numerical simulation of spacetime metric for gravitational wave simulations.

PUBLICATIONS

Puri, V. et al., *FLARE Decoder: Low-rank attention routing for causal language modeling* (In preparation).

Puri, V. et al., *FLARE: Fast Low-Rank Attention Routing Engine*. arXiv:2508.12594 (2025) (Under review).

Puri, V. et al., *Reduced order modeling with smooth neural fields*. JCP 2025, doi:10.1016/j.jcp.2025.113957.

S, V., Puri, V., et al., *Closure modeling for Burgers' turbulence*. MLST 2023, doi:10.1088/2632-2153/acb19c.

AWARDS

World Conference on Computational Mechanics | *Best poster in fluid dynamics* 2024

University of Illinois | *Theoretical and Applied Mechanics Merit Award* 2019

TECHNICAL SKILLS

ML Architectures: Efficient attention, causal transformers, memory-efficient prefill/decode, neural operators.

ML Systems: PyTorch internals, distributed training, mixed-precision training, Hugging Face.

GPU & Systems: Triton kernel development, FlashAttention-style block algorithms, online softmax, Nsight profiling.

Scientific Foundations: Numerical linear algebra, PDE-constrained modeling, finite elements, spectral methods.

Programming Languages: Python, Julia, C, Fortran 77, MATLAB, UNIX, L^AT_EX.